MUHAMMAD USAMA BIN AKHTAR KHAN

Applied AI Engineer · Senior Software Engineer | React · Next.js · TypeScript · Computer Vision · Local LLMs

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SUMMARY

Applied AI engineer and senior software engineer with 7+ years of production React, Next.js and TypeScript. Primary strengths include computer vision and local LLMs/TTS (YOLOPv2-Studio, KittenTTS, GPU inference for room visualisation), plus data-heavy admin panels and embeddable 3D tools. Leads frontend delivery and mentors engineers through component architecture, code review and quality tooling. Works across the stack (NestJS, Laravel, Python/FastAPI). Author of two open-source npm component libraries.

SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, Vue, Vite, Tailwind CSS, shadcn/ui, Three.js, Canvas, Web Components, TanStack Query, Redux, Zustand, React Hook Form + Zod, Material UI, Ant Design

Quality & accessibility: Vitest, Jest, Cypress, ESLint, Prettier, ARIA and keyboard patterns, regression QA harnesses

Backend & data: Node.js, NestJS, Python, FastAPI, Laravel (PHP), PostgreSQL, MariaDB, Prisma, Redis, Supabase, REST, GraphQL, WebSockets

Applied AI: SAM 3, Depth Anything V2, FLUX.2 [klein] inpainting (Nunchaku INT4), YOLOPv2, OpenCV, PyTorch, CUDA, Ollama

Mobile & delivery: Ionic / Capacitor (Android, iOS), Docker, GitHub Actions, Vercel

EXPERIENCE

Senior Software Engineer | FOCUS **Jul 2024 – Present**

Remote, Pakistan

- Lead frontend delivery for client products: component architecture, code review, quality tooling and mentoring engineers.
- **Giolindo** (React, Next.js, TypeScript): built a 32-frame Canvas 360° product viewer on my open-source viewer-360 library, plus a separate 3D/AR configurator where admins upload models and materials (textures stripped on save, materials applied at runtime).
- **Lumenweave** AI room visualiser: built the React/Vite + Three.js retailer embed (remove furniture from a room photo, place catalogue products at real scale, chained edits, undo, full-res export), backed by a FastAPI photo API and a separate authenticated GPU inference service (SAM 3, Depth Anything V2, FLUX.2 [klein] INT4 inpainting).
- Brought warm object removal to ~5–8 s on a single RTX 3060 (12 GB) with zero pixels changed outside the edit mask; prompt prewarming cut cold encoding from 13.6 s to 3.3 s; added an 11-room regression QA harness.
- **Global Foods**: delivered three Vue + Vite apps (customer, EPOS, operations) on Laravel/MariaDB with Reverb WebSockets and queue workers, including EPOS VAT, card payments and push notifications; Android and iOS builds via Ionic/Capacitor.
- **Herd WMS**: built a real-time warehouse management system with NestJS, PostgreSQL, Prisma, Redis and WebSockets.

Frontend Developer | Next Order **Jul 2023 – Jul 2024**

Remote, Australia

- Built the Next Order Control Panel in React and TypeScript for restaurant operations: menus, vouchers, inventory analytics, staff reports, service charges and store-level controls.
- Turned Figma designs into dashboards, data tables, forms, filters and reports using reusable Tailwind CSS components; integrated REST APIs with Axios, SWR and React Query.

Emumba **Sep 2020 – Jul 2023**

Islamabad, Pakistan · Frontend Developer (Jan 2023 – Jul 2023) · Markup Engineer (Sep 2020 – Jan 2023)

- Shipped fintech, fleet-tracking and analytics products, including Capital Bee (crypto tax and wallet analytics) and Flux (GPS fleet tracking).
- Built complex forms, data tables, charts and dashboards from Figma with React, Next.js, TypeScript, Redux, Formik, React Query, Material UI and Ant Design.

Software Engineer | TheTekSol — Pakistan **Sep 2018 – Sep 2020**

- Built a production review-management system used by COMSATS University Islamabad, Wah Campus, for final-year project and MS/PhD thesis evaluations (Laravel, jQuery, Bootstrap).
- Developed a point-of-sale system in React.js and its admin panel in Next.js.

PROJECTS

- **KittenTTS WebUI** — FastAPI + PyTorch local web UI for the KittenTTS 2 speech model: 47 voices, clip history, GPU with CPU fallback (UI MIT). github.com/usamakhangt4/kitten-tts-webui · Samples
- **git-auto-commit** — Rust CLI that reads staged diffs, drafts a commit message with a local Ollama model and confirms before committing; tests, installers and releases. github.com/usamakhangt4/git-auto-commit
- **shadcn-virtualized-tree** (npm) — typed headless + virtualized React tree for shadcn/ui with drag-and-drop, tri-state checkboxes and ARIA keyboard navigation. github.com/usamakhangt4/shadcn-virtualized-tree · Live demo
- **viewer-360** (npm) — TypeScript Canvas Web Component for 360° product rotation (drag, pinch/scroll zoom, high-res swap, network-aware loading) with React, Vue and Angular wrappers. github.com/usamakhangt4/viewer-360 · Live demo
- **YOLOPv2-Studio** — Python/Flask studio for YOLOPv2 object, lane and drivable-area detection with image and webcam input and automatic CUDA/MPS/CPU selection. github.com/usamakhangt4/YOLOPv2-Studio
- **Corpo Braindance** — local-first brief generator (Next.js 16, React 19, TypeScript, Tailwind CSS 4, Zustand); static export, 20 briefs in localStorage, no backend; reduced-motion and forced-colors support; Vitest and smoke tests. corpo-braindance.vercel.app

EDUCATION

BS Software Engineering — COMSATS University Islamabad, Wah Campus **2016 – 2020**